Latent Relationship Guided Stacked Sparse Autoencoder for Hyperspectral Imagery Classification

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***Abstract*— Classification is an important application of hyperspectral image (HSI). However, it is also a challenging research topic due to the spatial variability of spectral signature and limited training samples. To address these problems, a novel unsupervised feature learning method called latent relationship guided the stacked sparse autoencoder (LRSSAE) is developed in this article, which can effectively exploit the latent relationship under feature space to improve the ability of feature learning. Moreover, the superpixels constraint is employed on the feature representation to avoid the "salt-and-pepper" problem, and it is enforced on the latent relationship to enhance the latent relationship learning additionally. In LRSSAE, combining the stacked sparse autoencoder (SSAE) with the graph regularizations of latent relationship in each hidden layer and the superpixel constraints in the top layer, we extract feature representation in an unsupervised manner. And then, we present a customized iterative algorithm to optimize the LRSSAE. We evaluate the proposed method on three widely used HSI data sets comprehensively. The results demonstrate that our method achieves promising classification performance on these data sets and obtains improvements of 5.06%, 5.77%, and 2.11% in overall accuracy compared to the best SSAE method.**

***Index Terms*— Feature learning, hyperspectral image (HSI) classification, latent relationship, latent relationship guided the stacked sparse autoencoder (LRSSAE), superpixels constraint.**

NOMENCLATURE

A	Database images, $\mathbf{A} \in \mathbb{R}^{m_1 \times m_2 \times d}$.
X	Matrix of the original dataset, $\mathbf{X} \in \mathbb{R}^{M \times d}$.
Z	Total weighted sum of inputs.
a	Matrix of feature representation.
W	Weight matrix of the network.
b	Bias matrix of the network.
μ, α, ρ	They are used to control the regularizations in SAE.
G	Adaptive graph.
U	Weight matrix of **G**.
R	Latent relationship.
L	Laplacian matrix.
M	Number of image pixels.
L	Number of hidden layers.
S	Number of hidden units for each hidden layers.
$S^{(l)}$	Number of hidden units for lth layers.
ξ_1,ξ_2	They are used to balance the importance of the corresponding term.
K	Number of nearest neighbors.
T	Number of SAE model used in latent relationship guided the stacked sparse autoencoder (LRSSAE).
$\|\cdot\|_F$	Frobenius norm.
$\mathbf{Tr}(\cdot)$	Trace of matrix.

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I. INTRODUCTION

HYPERSPECTRAL imaging sensors can acquire continuous spectral information, ranging from the visible spectral to shortwave infrared bands and consecutive spatial data simultaneously. Compared with the RGB images, hyperspectral images (HSIs) can offer rich band information, which results in extensive applications [1]–[3]. One of the important applications is the HSI classification. However, it is difficult to achieve a good performance with the original HSI due to the curse of dimensionality, spatial variability of spectral signature and limited training samples [3]. Therefore, how to effectively reduce the dimensionality of HSI, overcome the spatial variability of spectral signature, and achieve higher classification performance with the limited training samples are becoming the important tasks for HSI classification.

A lot of works have been developed to address these problems [4]–[8]. Among these HSI classification methods, a major finding is that suitable feature representation can make a classification model work well [9]. Therefore, feature

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representation plays a key role in the HSI classification. According to data label, these works can be divided into three categories: supervised, semi-supervised, and unsupervised feature learning. Supervised feature learning methods usually achieve high accuracies with a large amount of labeled data, e.g., dictionary learning [10], sparse representation [11], and support vector machine (SVM) [12], [13]. Since it is time-demanding and labor-intensive for acquiring the annotated data, supervised feature learning is more challenging. In comparison with supervised feature learning methods, the semi-supervised and unsupervised feature learning methods have more practical applications in HSI analysis, since less labeled data are required in them. Semi-supervised methods also extract feature representations in a supervised manner but use smaller-sized labeled samples than the supervised methods [14]–[17], whereas for unsupervised methods, these do not require labeled data. Principle component analysis (PCA) [18], independent component analysis (ICA) [19], and LDA-based methods [20]–[22] have been utilized to extract linear feature representation in an unsupervised manner extensively. There are also diverse nonlinear unsupervised feature learning methods [23]–[25]. However, these conventional methods have limited abilities to handle nonlinear data, while HSI often suffers from inherently nonlinear issues [26].

In recent years, deep learning models (DLMs) have emerged as one promising method in many computer vision applications. Meanwhile, some of them have been employed for the HSI analysis successfully. Autoencoder (AE), deep belief network (DBN), and convolutional neural network (CNN) are widely employed to extract nonlinear feature representation. AE as a significant branch of the DLMs, which can learn compressed feature representation in an unsupervised manner, has attracted considerable interests [27], [28]. However, the ability of basic AE is limited, which cannot well handle the issue when the number of hidden units is more than the dimension of input data. To address this problem, sparse AE (SAE) in which the sparsity is integrated into the conventional AE model to learn more efficient sparse feature representations has been developed [29].

Commonly, the basic SAE model merely learns pixel-level feature representation, ignoring spatial correlation among image pixels. It is often easily affected by noise and cannot obtain high classification accuracy. Thus, a lot of approaches have been proposed to tackle this issue [30]–[34]. The method proposed by Chen *et al.* [3], which extracted spatial correlation from PCA transformation based on the AE model, obtained a better classification performance than the one that used only spectral or spatial information. Similarly, Abdi *et al.* [28] also extracted the spectral and spatial features simultaneously, fed it into the SAE model, and obtained a high HSI classification accuracy. Chao *et al.* [35] established a multiscale spatial feature learning procedure by feeding the SAE model with multiscale image patches. Ma *et al.* [36] combined a collaborative representation including spatial correlation and spectral information, and a spatial regularization was added in the energy function to improve the performance of the deep AE model. Most of the existing methods focus on extracting spatial correlation from HSI image pixels, while ignoring the latent relationship existing in feature space. Wang *et al.* [37] proposed the LRAGE to learn the latent relationship with manifold learning algorithms, which achieved a better performance in aerial scene classification. As indicated in [37], the appropriate latent relationship can be beneficial for improving the ability of feature learning. Thus, the latent relationship between pixels can also be adopted to learn the more representative and discriminative HSI features with manifold learning in our model.

The accuracy of feature representation depends not only on the latent relationship learning but also on the constraint of the semantic information around pixels. Pixel-based classification methods often lead to the "salt-and-pepper" problem in HSI classification due to lack of contextual information among pixels and contain noise and mixed pixel in HSI images [38]. The superpixel is usually generated by a group of surrounding pixels. In one superpixel, pixels should have similar image features and the same labels. Then the superpixels can provide contextual semantic information among pixels. Thus, superpixel constraints can be employed to preserve the semantic information in HSI. Superpixel-based feature learning methods have been demonstrated as an effective way to enhance the HSI classification performance. In [39], a superpixel map was obtained by using downsampling method, which guided a framework to learn sparse representations for improving HSI classification accuracy. Mei *et al.* [40] presented a subspace learning method considering pixel-level and superpixel-level information. In pixel-level, label space, spectral information, and spatial correlation were used to extract pixel-based features, whereas in superpixel-level, generated superpixels served as the feedback information to the subspace learning. To avoid the "salt-and-pepper" problem and improve the HSI classification accuracy, a superpixel constraint was embedded into the low-rank representation (LRR) in [41], which achieved a better performance in HSI classification than original LRR. In our model, the generated superpixels take the latent relationship and feature representations into account simultaneously.

Under the above considerations, it is important to exploit an efficient and effective feature learning approach, which can adopt the latent relationship between HSI pixels and the advantages of superpixels to improve the accuracy of HSI classification. Since DLMs have the ability to extract hierarchical features and yielding high classification accuracy than those traditional methods, we integrate the latent relationship between a pixel and the superpixel constraints into the stacked SAE (SSAE) model. In this framework, the latent relationship is integrated into each hidden layer of SSAE to guide feature learning. Furthermore, the superpixel constraints, including the latent relationship and the learned pixels' features are defined in the top layer, are also employed to improve the ability of feature learning. The objective function of LRSSAE is constructed with the SAE and the superpixel constraints. Finally, a customized iterative algorithm is introduced to optimize the LRSSAE. In the experiment, we test our proposed method on three extensively used HSI classification data sets, and the experimental results show that the classification performance obtained from our approach can exceed most of unsupervised HSI classification methods.

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The main contributions of this article are summarized as follows.

1) The latent relationship between image pixels is constructed and embedded into the network of LRSSAE with the manifold assumption, which can enhance the HSI feature learning ability of our method.
2) Superpixels are utilized to avoid the "salt-and-pepper" problem, provide feedback information to the latent relationship learning, and boost the accuracies of feature learning and latent relationship learning further.
3) The loss functions constructed by superpixels are integrated into the objective function of SAE. We develop a customized iterative algorithm to optimize the LRSSAE. And the experiment results validate that the method proposed in this article is very effective and efficient for HSI classification.

For clarity, we illustrate important notation and definitions used throughout this article in the Nomenclature.

II. Related Works

In this section, we briefly review the related studies including SAE and manifold learning.

A. SAE

The AE is a symmetrical neural network containing the input, hidden, and reconstruction layers, which can be utilized to automatically learn abstract features in an unsupervised manner by minimizing the discrepancy between the raw input and its reconstruction [42].

In general, a basic AE consists of two parts: the encoder and the decoder. The encoder maps the input vector $\mathbf{x} \in \mathbb{R}^{\mathrm{M}}$ to an abstract feature representation $\mathbf{a} \in \mathbb{R}^{S}$. The new feature representation is then used to reconstruct an approximation vector $\mathbf{y} \in \mathbb{R}^{\mathrm{M}}$ of the input vector via a decoder. The procedures can be shown as

$$\begin{aligned} \mathbf{a} &= f(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) \\ \mathbf{y} &= f(\mathbf{W}_2\mathbf{a} + \mathbf{b}_2) \end{aligned} \tag{1}$$

where $\mathbf{W}_1 \in \mathbb{R}^{S\times M}$, $\mathbf{W}_2 \in \mathbb{R}^{M\times S}$, $\mathbf{b}_1 \in \mathbb{R}^{S\times 1}$, $\mathbf{b}_2 \in \mathbb{R}^{M\times 1}$ represent the weights and biases, respectively. And f represents the activation function, usually it is a nonlinear sigmoidal function.

Commonly, the optimization function of the AE can be shown in the following equation:

$$J_{\mathrm{AE}}(\mathbf{W},\mathbf{b}) = \frac{1}{\mathrm{M}}\sum_{i=1}^{M}\left(\frac{1}{2}\|\mathbf{y}_i - \mathbf{x}_i\|_2^2\right) + \frac{\mu}{2}\sum_{l=1}^{L}\sum_{i=1}^{S^{(l)}}\sum_{j=1}^{S^{(l+1)}}\left(\mathbf{W}_{ij}{}^{(l)}\right)^2 \tag{2}$$

where the first term denotes the reconstruction error term, which is optimized by minimizing the reconstruction error between raw input and reconstructed vector. The second term denotes the weight decay term, which is employed to prevent overfitting, μ is the weight decay parameter, and the number of layer L is set 2 for the standard AE.

The SAE imposes a sparse constraint on the hidden layer to achieve an "overcomplete" nonlinear mapping of the input vector, in other words, the dimension of hidden units can be larger than the dimension of input [42]. Therefore, the optimization function of SAE can be defined as

$$J_{\mathrm{SAE}}(\mathbf{W},\mathbf{b}) = J_{\mathrm{AE}}(\mathbf{W},\mathbf{b}) + \alpha\sum_{j=1}^{S}\mathrm{KL}(\rho\|\hat{\rho}_j) \tag{3}$$

where $\mathrm{J_{AE}}(\mathbf{W},\ \mathbf{b})$ is defined previously, and the new term is sparse penalty term that can be optimized by minimizing the Kullback–Leibler (KL) divergence between $\hat{\rho}_j$ and ρ, and α is the sparse penalty parameter which controls the weight of the sparse penalty term. The KL divergence can be defined as

$$\mathrm{KL}(\rho\|\hat{\rho}_j) = \sum_{j=1}^{S}\left(\rho\log\frac{\rho}{\hat{\rho}_j} + (1-\rho)\log\frac{1-\rho}{1-\hat{\rho}_j}\right) \tag{4}$$

where $\hat{\rho}_j = (1/S)\sum_{j=1}^{S}\mathbf{a}_j$, which is the average activation of hidden unit j over the training data $\mathbf{x}$, and ρ is the target average activation of hidden unit j, that is a very small value close to 0, commonly.

Basically, the purpose of training the SAE is to find optimal parameters $\theta = \{\mathbf{W},\mathbf{b}\}$ by minimizing the objective function. The minimization procedure can be performed via the stochastic gradient descent and back-propagation algorithm, iteratively [43], [44].

B. Manifold Learning

The local intrinsic geometry structure of the data manifold is ignored in most existing subspace clustering methods [45]. The manifold learning method is utilized to discover the underlying intrinsic geometry of the data manifold since it assumes that the intrinsic geometry of data lies in a low-dimensional manifold in a high-dimensional space. There are many prominent manifold learning algorithms. We adopt the Laplacian Eigenmap method to model the local geometric structure of the data manifold by building a nearest-neighbor graph [46], [47]. For data set $\mathbf{X}$, a nearest-neighbor graph can be constructed by the neighborhood information of the data, each data point serves as a node on the nearest-neighbor graph, and the proximity of neighboring data points is employed to determine connectivity between different nodes.

Under the manifold assumption, the latent structure of the original data can be discovered in the low-dimensional representation. If two data points $\mathbf{X}_i$ and $\mathbf{X}_j$ are compact in the intrinsic geometry of the data distribution, their corresponding representations $\mathbf{h}_i$ and $\mathbf{h}_j$ should also be compact to each other [48]. The corresponding objective function can be defined as:

$$\Lambda = \frac{1}{2}\sum_{i,j=1}^{S}\mathbf{V}_{ij}\|(\mathbf{h})_i - (\mathbf{h})_j\|_2^2 = \mathrm{tr}(\mathbf{h}\mathbf{L}^*\mathbf{h}^{\mathbf{T}}) \tag{5}$$

where $\mathbf{V}_{ij}$ denotes weight matrix and $\mathbf{L}^*$ denotes the Laplacian matrix.

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III. Proposed Method

In this section, a novel feature learning method LRSSAE is introduced. First, the motivation of this article is presented in Section III-A. Second, the SSAE model is summarized in Section III-B. Next, the construction of the latent relationship is described in Section III-C and then the objective function of LRSSAE is described in Section III-D. At last, the optimization of the proposed model LRSSAE is given in Section III-E.

A. Motivation

HSI classification suffers from the noise and high correlation between spectral bands. Therefore, an efficient feature learning method is necessary to develop for improving the HSI classification accuracy. SAE can extract nonlinear features in an unsupervised manner, which has been widely used in HSI classification. In this article, we adopt the SSAE for hyperspectral classification, since the SSAE can extract more abstract and higher-order features of data [49], [50]. However, the basic SSAE model only employs the pixel-level spectral information for HSI classification. Overlooking the latent relationship and contextual information among pixels, the HSI classification accuracy is limited. Thus, in our proposed method LRSSAE, the latent relationship is integrated into the SSAE model to improve the accuracy of feature learning. In addition, the constraint from superpixels is introduced to overcome the "salt-and-pepper" problem and further enhance the latent relationship learning. Under manifold learning assumption, the latent relationship is embedded into the framework of feature learning, aiming to adopt the latent relationship to guide the HSI feature learning. Moreover, the developed framework of feature learning is optimized with a customized iterative algorithm.

B. SSAE

Inspired by the DLMs which automatically learn comprehensive high-level features from data by stacking unsupervised modules, we adopt SSAE that is developed via stacking SAE layer-by-layer, the hidden layer of each SAE is wired to the input layer of subsequent SAE, and then removing the reconstruction layer of each SAE, finally stacking up the hidden layers to construct a new network—SSAE.

Given a 3-D hyperspectral data set $\mathbf{A} \in \mathbb{R}^{m_1 \times m_2 \times d}$, where $m_1 \times m_2$ denotes the number of HSI pixels and d denotes the number of bands, we put the data set $\mathbf{A}$ into a matrix $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_M] \in \mathbb{R}^{M \times d}$, where $M = m_1 \times m_2$. Assuming there are L layers in the SSAE framework, which are employed to automatically extract different order features. For the first SAE, the input is an HSI pixel vector X_i, the primary representation of the hidden layer is $\mathbf{a}^{(1)}$, which can be computed as: $\mathbf{a}^{(1)} = f(\mathbf{z}^{(1)}) = f(\mathbf{W}^{(1)}\mathbf{X} + \mathbf{b}^{(1)})$, where $\mathbf{W}^{(1)}$ denotes the weight matrix and $\mathbf{b}^{(1)}$ denotes the bias vector, which are learned by the first SAE, f is a nonlinear sigmoid function. Next, the primary representation $\mathbf{a}^{(1)}$ serves as the input of the next SAE to learn the secondary representation $\mathbf{a}^{(2)}$, the procedure can be defined as: $\mathbf{a}^{(2)} = f(\mathbf{z}^{(2)}) = f(\mathbf{W}^{(2)}\mathbf{a}^{(1)} + \mathbf{b}^{(2)})$, where the weight matrix $\mathbf{W}^{(2)}$ and bias vector $\mathbf{b}^{(2)}$ are learned in the second SAE. Similarly, the hidden layer of the lth ($1 \le l \le L$) SAE can be computed as: $\mathbf{a}^{(l)} = f(\mathbf{z}^{(l)}) = f(\mathbf{W}^{(l)}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)})$, where $\mathbf{W}^{(l)}$ and $\mathbf{b}^{(l)}$ are learned in the lth SAE, which can extract more abstract and higher-order feature representations.

C. Latent Relationship Construction

The latent relationship should be relatively stable no matter how the feature of pixels changes, which used to measure the similarity of HSI pixels. Therefore, to obtain accurate feature representation, the latent relationship $\mathbf{R}$ constructed by a graph is introduced to enhance the feature learning ability. In addition to the latent relationship, the spectral–spatial similarity can also be used to measure the relationship between HSI pixels. A graph-based method is employed to construct the spectral–spatial similarity. In the graph, each vertex corresponds to one image pixel vector $\mathbf{X}_i$, and a weight matrix $\mathbf{U}$ measures the correlation between two different samples to seek the nearest neighbors. We construct the spectral graph $\mathbf{G}_1$ to describe the similarity among the spectral information of feature representations and the spatial graph $\mathbf{G}_2$ to describe the correlation among spatial information of feature representations, respectively. K-nearest-neighbor (KNN) algorithm is employed to choose the neighbor pixels. The weight matrix $\mathbf{U}_1$ of the spectral graph can be utilized to determine the similarity among neighbor pixels. The weight matrix $\mathbf{U}_1$ is defined as

$$\mathbf{U}_1(i,j) = \begin{cases} \exp\left(-\mathbf{r}_{ij}{}^2\right), & x_i \in \mathcal{N}_{k_1}(x_j) \text{ or } x_j \in \mathcal{N}_{k_1}(x_i) \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where $\mathbf{r}_{ij} = \|\mathbf{a}_i{}^{(l)} - \mathbf{a}_j{}^{(l)}\|_2$, $\mathbf{a}_j{}^{(l)}$ denotes the feature representation of $\mathbf{X}_j$ in the lth layer, and $\mathcal{N}_{k_1}(x_j)$ denotes the KNNs of the pixel $\mathbf{X}_j$.

In the spatial graph, the weight matrix $\mathbf{U}_2$ is constructed by the Euclidean distance of different features and defined by the following function:

$$\mathbf{U}_2(i,j) = \begin{cases} \exp\left(-\frac{\|\mathbf{P}_i - \mathbf{P}_j\|_2^2}{\gamma_1}\right), & x_i \in \mathcal{N}_{k_2}(x_j) \text{ or } x_j \in \mathcal{N}_{k_2}(x_i) \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where γ_1 is a trade-off parameter, $\|P_i - P_j\|$ denotes the spatial distance of the KNNs between the low-dimensional features, $\mathcal{N}_{k_2}(x_j)$ denotes the KNNs of $\mathbf{X}_j$.

We combine the exploited correlation in spectral with spatial to represent the similarity of HSI pixels, which is denoted by $\mathbf{U}$. Theoretically, latent relationship should be equal to $\mathbf{U}$, so we constructed (8). The latent relationship should approximate the $\mathbf{U}$ established by the accurate feature representation

$$\begin{aligned} &\mathbf{U} = \mathbf{U}_1 + \mathbf{U}_2 \\ &\text{s.t. } \mathbf{R} = \mathbf{U}. \end{aligned} \tag{8}$$

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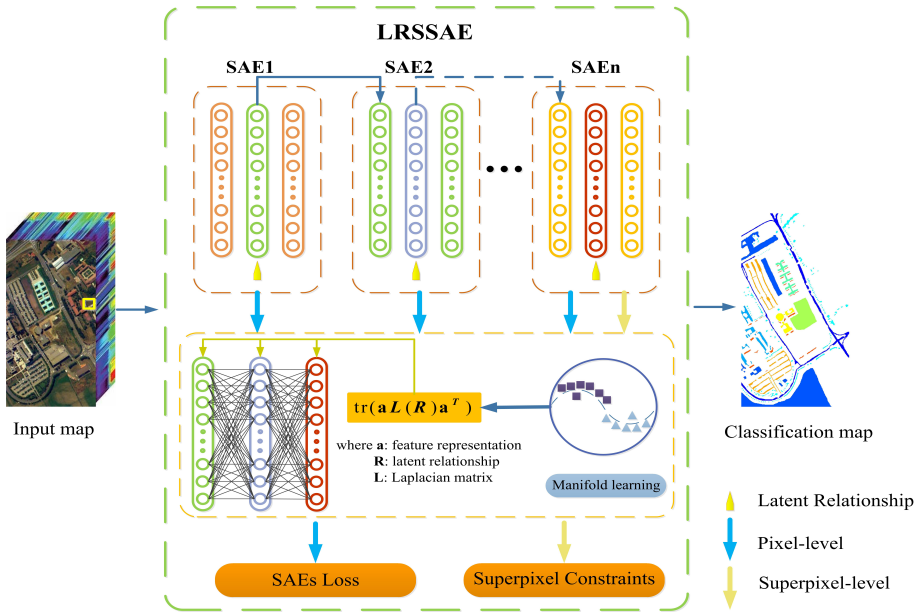

Fig. 1. Framework of LRSSAE. To boost the feature learning performance, the latent relationship constructed by the manifold learning method is embedded into the SAE model and the superpixel-level constraints provide feedback information to the latent relationship learning and feature learning. Thus, there are two losses in the proposed method, including SAEs loss and superpixel constraints.

D. Latent Relationship Guided SSAE

A standard SSAE only considers the spectral information of a single pixel and ignores the locally smooth nature in feature learning, i.e., the latent relationship of feature representations should be consistent with the data structure of HSI, which can improve the ability of feature learning and extract more representative feature representation than the basic model. Thus, in LRSSAE, the manifold learning algorithm is employed to learn the latent relationship, which is used for guiding the feature learning. Additionally, to prevent the degradation of classification performance from the "salt-and-pepper" problem, a superpixel constraint is integrated into the objective function of SAE. The LRSSAE framework is shown in Fig. 1. Considering locally smooth in the feature space and contextual semantic information around pixels, a latent relationship learning and a superpixes regularization can be integrated into the SAE. The objective function can be defined as follows:

$$\begin{aligned} J(\theta, \mathbf{R}) = &\sum_{t=1}^{T-1} \left(J_{\mathrm{SAE}}(\theta) + \xi_1 J_{\mathrm{LR}}(\theta, \mathbf{R})\right) \\ &+ \sum_{t=T} \left(J_{\mathrm{SAE}}(\theta) + \xi_1 J_{\mathrm{LR}}(\theta, \mathbf{R}) + \xi_2 J_{\mathrm{SP}}(\theta, \mathbf{R})\right) \\ \text{s.t. } &\mathbf{R} = \mathbf{U}(\theta) \end{aligned} \tag{9}$$

where θ denotes the collection of weight matrix $\mathbf{W}$ and biases vector $\mathbf{b}$. T denotes the number of SAE used in LRSSAE model, ξ_1 and ξ_2 are the trade-off factors. J_{SAE} denotes the SAE objective function, which is defined in (3). J_{LR} denotes the regularization term of the latent relationship, which is constructed in each layer of SSAE. And the superpixels regularization term is represented by J_{SP}, which is used only in the top layer of SSAE.

In (9), the corresponding term ξ_1 is employed to optimize the latent relationship. Manifold learning is used to learn the latent relationship, and the latent relationship should meet the manifold assumption. If the HSI pixels are close in latent relationship, they have similar data structures in a low-dimensional feature. Under the assumption, the corresponding objective function is expressed as

$$\begin{aligned} J_{\mathrm{LR}} &= \frac{1}{2} \sum_{i,j}^{S^{(l)}} \mathbf{R} \left\| (\mathbf{a})_i^{(l)} - (\mathbf{a})_j^{(l)} \right\|_F^2 \\ &= \mathrm{tr}(\mathbf{a}^{(l)} \mathbf{L} (\mathbf{a}^{(l)})^T \end{aligned} \tag{10}$$

where the Laplacian matrix of latent relationship graph can be denoted by $\mathbf{L}$, which is constructed by $\mathbf{L} = \mathbf{D} - \mathbf{R}$. $\mathbf{D}$ is a diagonal matrix, in which the diagonal element (i, i) equals the sum of the ith row of $\mathbf{R}$.

The corresponding term of ξ_2 is used to address the "salt-and-pepper" problem. The contextual information around pixels is extracted in superpixels under feature space. Numerous approaches have been proposed for superpixels generation. We adopt the adaptive mean shift-clustering method to generate the superpixels [51]. Considering the feature representations extracted by the top layer of SSAE is utilized to classify, we integrate the superpixels regularization on the top layer to optimize the feature learning and latent relationship learning. Therefore, not only the feature representation but also the latent relationship should satisfy the following assumptions.

1) Pixels in the same superpixel group should have similar data structures.
2) Constructed the latent relationship in the same superpixel group should be consistent with each other.

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Under these assumptions, if jth pixel belongs to kth superpixel, the constraint function can be defined as follows:

$$J_{\mathrm{SP}} = \sum_{j=1}^{S^{(l)}} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2}\right) + \varphi\|\mathbf{R} - \bar{\mathbf{R}}\|_F^2 \tag{11}$$

where $\mathbf{a}_j$ denotes the feature representations, ς_k represents the feature of the kth superpixel, and γ_2 denotes the heat kernel. φ is a trade-off parameter and $\bar{\mathbf{R}}$ is the statistical latent relationship of superpixel k.

E. Optimization of LRSSAE Model

The optimization of LRSSAE is summarized in algorithm 1. We employed the greedy layer-wise approach for training the LRSSAE by training each layer in a sequential manner. The optimization procedure in each SAE is similar except the top layer. The objective function of single SAE can be defined as

$$\begin{aligned} J(\theta, \mathbf{R}) &= J_{\mathrm{SAE}}(\theta) + \xi_1 J_{\mathrm{LR}}(\theta, \mathbf{R}) + \xi_2 J_{\mathrm{SP}}(\theta, \mathbf{R}) \\ &= \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2}\|\mathbf{y}_i - \mathbf{x}_i\|_2^2\right) + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)}\right)^2 \\ &\quad + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho\|\hat{\rho}_j) + \xi_1 \operatorname{tr}(\mathbf{aLa}^T) \\ &\quad + \xi_2 \left(\sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2}\right) + \varphi\|\mathbf{R} - \bar{\mathbf{R}}\|_F^2\right) \\ &\quad s.t.\ \mathbf{R} = \mathbf{U}(\theta). \end{aligned} \tag{12}$$

The linearized alternating direction method with adaptive penalty (LADMAP) [52] is used to solve (12), the augmented Lagrangian function of (12) is

$$\begin{aligned} &J(\theta, \mathbf{R}, \omega, \tau) \\ &= \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2}\|\mathbf{y}_i - \mathbf{x}_i\|_2^2\right) + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)}\right)^2 \\ &\quad + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho\|\hat{\rho}_j) + \xi_1 \operatorname{tr}(\mathbf{aLa}^T) \\ &\quad + \xi_2 \left(\sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2}\right) + \varphi\|\mathbf{R} - \bar{\mathbf{R}}\|_F^2\right) \\ &\quad + \langle \omega,\ \mathbf{R} - \mathbf{U}(\theta)\rangle + \frac{\tau}{2}\|\mathbf{R} - \mathbf{U}(\theta)\|_F^2 \\ &= \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2}\|\mathbf{y}_i - \mathbf{x}_i\|_2^2\right) + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)}\right)^2 \\ &\quad + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho\|\hat{\rho}_j) + \xi_1 \operatorname{tr}(\mathbf{aLa}^T) \\ &\quad + \xi_2 \left(\sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2}\right) + \varphi\|\mathbf{R} - \bar{\mathbf{R}}\|_F^2\right) \\ &\quad + \frac{\tau}{2}\left\|\mathbf{R} - \mathbf{U}(\theta) + \frac{\omega}{\tau}\right\|_F^2 - \frac{1}{2\tau}\|\omega\|_F^2 \end{aligned} \tag{13}$$

where $\omega \in \mathbb{R}^{S\times S}$ is the Largrange multiplier and τ is the penalty parameter.

θ is solved when $\mathbf{R}$ are fixed. The optimization function can be expressed as follows:

$$\begin{aligned} \min_{\theta} J(\theta) &= \min_{\theta} \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2}\|\mathbf{y}_i - \mathbf{x}_i\|_2^2\right) \\ &\quad + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)}\right)^2 \\ &\quad + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho\|\hat{\rho}_j) + \xi_1 \operatorname{tr}(\mathbf{aLa}^T) \\ &\quad + \xi_2 \sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2}\right) \\ &\quad + \frac{\tau}{2}\left\|\mathbf{U}(\theta) - \left(\mathbf{R} + \frac{\omega}{\tau}\right)\right\|_F^2. \end{aligned} \tag{14}$$

$\mathbf{R}$ is solved when θ are fixed. The optimization function defined in (13) is expressed as follows:

$$\begin{aligned} \min_{\mathbf{R}} J(\mathbf{R}) &= \min_{\mathbf{R}} \xi_1 \operatorname{tr}(\mathbf{aLa}^T) + \xi_2\varphi\|\mathbf{R} - \bar{\mathbf{R}}\|_F^2 \\ &\quad + \frac{\tau}{2}\left\|\mathbf{R} - \left(\mathbf{U}(\theta) - \frac{\omega}{\tau}\right)\right\|_F^2. \end{aligned} \tag{15}$$

The Largrange multiplier is updated as follows:

$$(\omega)_{\mathrm{new}} = (\omega)_{\mathrm{old}} + \tau(\mathbf{R} - \mathbf{U}(\theta)). \tag{16}$$

More details of LRSSAE optimization can be referred to the Appendix.

Algorithm 1 LRSSAE

Input: Input data set $\mathbf{X}$, the number of SAE $\mathbf{T}$, parameters μ, α, ρ, ξ_1, ξ_2;
Initialize: θ, $\mathbf{R}$, ω, $t = 1$; τ=1×10^4, $\tau_{\max}$=1×10^{10}, $\lambda = 1.2$, σ=1×10^{-3};
Procedure:
1:**While** $t \le T$
  1.1: **While** stopping criterion has not been met **do**
  1.2: Update θ and $\mathbf{R}$ by solving Eq. (14) and (15);
  1.3: Update the Lagrange multipliers by Eq. (16);
  1.4: Update the penalty parameter τ by
    $\tau = \min(\lambda\tau, \max\tau)$;
  1.5: Check the criterion:
    $\|\mathbf{R} - \mathbf{U}\|_F \le \sigma$
  1.6: End while
  Output: the matrix θ and $\mathbf{R}$.
2:**Update**: $t = t + 1$
3: End while
Output: the matrix θ for each SAE

IV. Experimental Results

In this section, we evaluate the performance of the LRSSAE method for classification on three widely used HSI data sets. Afterward, we compare the classification results of the LRSSAE with the related methods.

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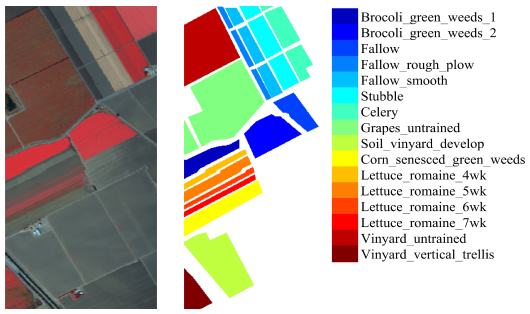
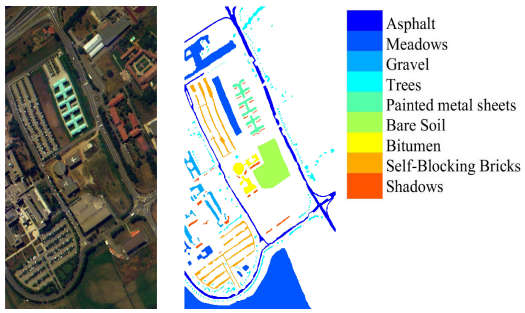


(a) (b)

Fig. 2. Salinas data set. (a) False RGB image. (b) Ground-truth map.



(a) (b)

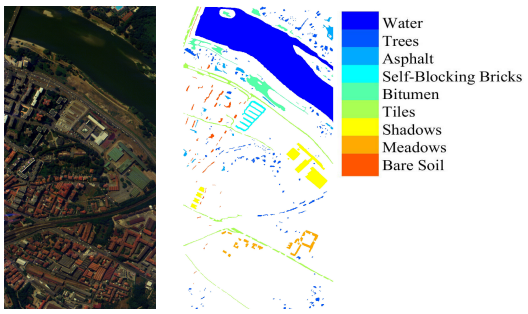
Fig. 3. PaviaU data set. (a) False RGB image. (b) Ground-truth map.

A. Data Sets Description and Experiment Design

We test the proposed algorithm on three benchmark HSI data sets.

1) The Salinas data set, which was acquired by AVIRIS over Salinas Valley, CA, USA. It consists of 512 × 217 pixels and contains 224 spectral bands ranging from the visible light to SWIR, with high spatial resolution 3.7 m. There are 204 bands (removing 20 water absorption bands) used in this experiment. The Salinas contains 54 129 labeled pixels totally and 16 classes, including vegetables, bare soils, and vineyard fields. The false RGB image and ground-truth map are shown in Fig. 2.
2) The other two data sets, the University of Pavia (PaviaU) and the centre of Pavia (PaviaC), which were gathered by the ROSIS over Pavia in northern Italy. PaviaU consists of 610 × 340 pixels and includes 103 spectral bands. There are 1096 × 492 pixels and 102 spectral bands in the PaviaC. The spatial resolution is 1.3 m. These data sets contain nine different classes, respectively. Figs. 3 and 4 show the false RGB image and ground-truth map.

In this experiment, the HSI classification can be divided into two steps: pretrained the feature extraction model in an unsupervised manner, and trained softmax classifier using little labeled data samples and finetune the model parameters further. Thus, we organize our experiments to validate the proposed method. First, the LRSSAE model is pretrained by the greedy layer-wise method. We randomly select 200 samples of each category as the training samples and the remaining samples as the testing samples to learn feature representation. And then for the Softmax classifier, we randomly select



(a) (b)

Fig. 4. PaviaC data set. (a) False RGB image. (b) Ground-truth map.

TABLE I
LAND COVER TYPES WITH SAMPLES' NUMBER FOR THE SALINAS DATA SET

Class	Land Cover Type	Training samples	Testing samples	Labeled data (2%)
1	Brocoli_green_weeds_1	200	1809	40
2	Brocoli_green_weeds_2	200	3526	74
3	Fallow	200	1776	39
4	Fallow_rough_plow	200	1194	27
5	Fallow_smooth	200	2478	53
6	Stubble	200	3759	79
7	Celery	200	3379	71
8	Grapes_untrained	200	11071	225
9	Soil_vinyard_develop	200	6003	124
10	Corn_senesced_green_weeds	200	3078	65
11	Lettuce_romaine_4wk	200	868	21
12	Lettuce_romaine_5wk	200	1727	38
13	Lettuce_romaine_6wk	200	716	18
14	Lettuce_romaine_7wk	200	870	21
15	Vinyard_untrained	200	7068	145
16	Vinyard_vertical_trellis	200	1607	36
	Total	3200	50929	1076

c samples with label to train the classifier and others are unlabeled data to test the classifier. For Salinas and PaviaU, c is set to 2%; while in PaviaC data set c is set to 1%, since there are many image pixels in those data set. The detailed information of test samples and training samples used in feature learning and labeled data used in classifier, which can be seen in Tables I–III.

B. Comparisons With Related Approaches

Since the proposed LRSSAE model is an unsupervised feature representation learning method, we compare it with the following related approaches by HSI classification accuracy.

1) *KNNs:* Using the Euclidean distance to measure the similarity of image pixels and a Gaussian kernel to reweight the edges.
2) *Kernel-PCA (KPCA) [53]:* It can be considered as a nonlinear PCA, which uses a Gaussian kernel function to obtain feature representations.
3) *Local Linear Embedding (LLE) [46]:* It reconstructs each data through linear weights which obtained from the selected neighbors, and then finds the best reconstructed low-dimensional data as the feature representation.
4) *Robust Subspace Segmentation by LRR [54]:* LRR represents the data with a linear combination of the

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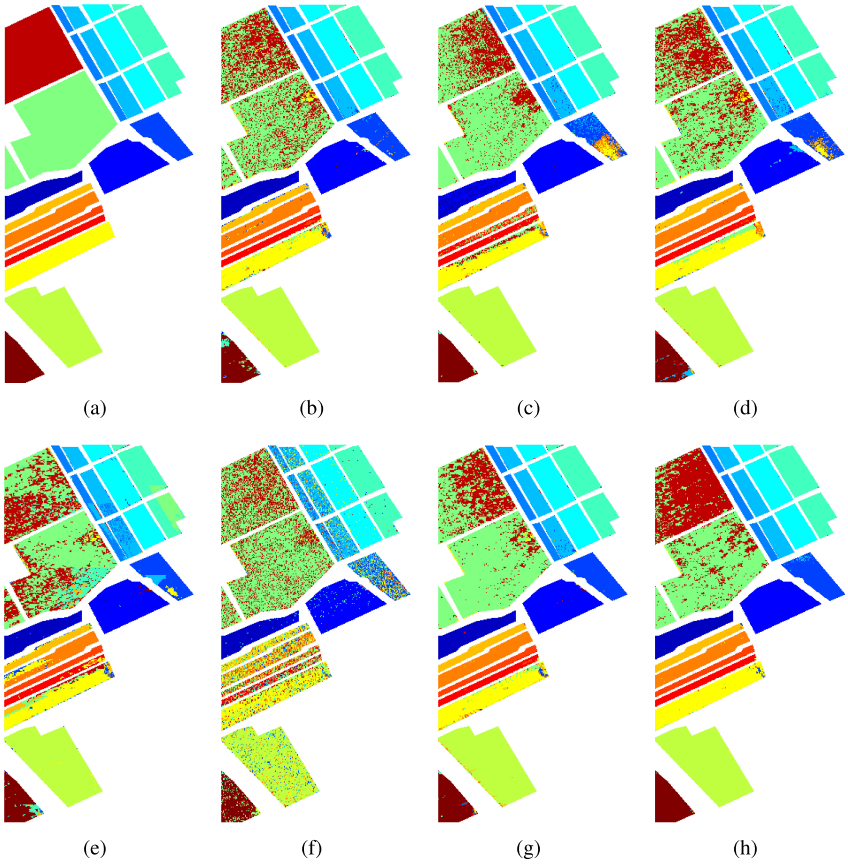

Fig. 5. Classification maps of the Salinas data set by selecting 2% labeled samples for each class. Classification map obtained by (a) ground truth, (b) KNN, (c) KPCA, (d) RICA, (e) LLE, (f) LRR, (g) SSAE, and (h) LRSSAE.

TABLE II
LAND COVER TYPES WITH SAMPLES' NUMBER FOR THE UNIVERSITY OF PAVIA DATA SET

Class	Land Cover Type	Training samples	Testing samples	Labeled data (2%)
1	Asphalt	200	6431	132
2	Meadows	200	18449	372
3	Gravel	200	1899	41
4	Trees	200	2864	61
5	Painted metal sheets	200	1145	26
6	Bare Soil	200	4829	100
7	Bitumen	200	1130	26
8	Self-Blocking Bricks	200	3482	73
9	Shadows	200	747	18
	Total	1800	40976	849

TABLE III
LAND COVER TYPES WITH SAMPLES' NUMBER FOR THE CENTRE OF PAVIA DATA SET

Class	Land Cover Type	Training samples	Testing samples	Labeled data (1%)
1	Water	200	65078	652
2	Trees	200	6308	65
3	Asphalt	200	2705	29
4	Self-Blocking Bricks	200	1940	21
5	Bitumen	200	6349	65
6	Tiles	200	7385	75
7	Shadows	200	7087	72
8	Meadows	200	2922	31
9	Bare Soil	200	1965	21
	Total	1800	101739	1031

bases in the given dictionary. It seeks the lowest-rank representation of all data jointly.

5) *Reconstruction ICA (RICA) [55]:* It uses a robust soft reconstruction cost for ICA, instead of the orthonormality constraint, which can learn overcomplete sparse features in an unsupervised manner.
6) *SSAE:* It is constructed by stacking the shallow SAE, which can extract the sparse features in the deeper network.

The above six approaches extract feature representation in an unsupervised manner. In KNN and LLE, the number of nearest neighbors must be specified, which is selected from the set {3, 4, 5, 6, 7, 8, 9, 10}. For KNN, LLE, and KPCA, using the kernel-based distance to measure the distance between pixel features. It is computed by $\exp(-(\|X_i - X_j\|^2)/\sigma)$, where σ denotes the heat kernel, and it is selected from the set $\{10^{-9}, 10^{-8}, \ldots, 10^{8}, 10^{9}\}$. And the number of dimensions for embedding is selected from the set {30, 40, 50, 60, 70, 80, 90, 100} in KNN, KPCA, LLE, and RICA. Applying the given number of nearest number and distance measurement from the defined graph function in LRR.

To validate the effectiveness of our proposed method, we compared the basic SSAE and LRSSAE method additionally. In LRSSAE, the graph-based method is used, not only in the latent relationship but also in the spectral and spatial correlation that can be employed to constrain the latent relationship

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TABLE IV

HSI CLASSIFICATION RESULTS BY SELECTING 2% LABELED SAMPLES FOR EACH CLASS ON THE SALINAS DATA SET

Land Cover Type	KNN	KPCA	RICA	LLE	LRR	SSAE	LRSSAE
Brocoli_green_weeds_1	99.99	48.18	99.35	86.65	86.66	98.55	99.01
Brocoli_green_weeds_2	97.69	99.78	97.12	95.27	95.97	97.28	99.81
Fallow	90.78	57.99	82.74	62.95	37.75	72.67	98.74
Fallow_rough_plow	97.53	95.76	99.35	72.02	40.82	99.43	99.35
Fallow_smooth	96.88	96.75	94.47	64.00	66.39	98.61	98.87
Stubble	99.77	99.77	98.66	84.68	96.21	99.79	99.62
Celery	96.52	99.33	99.07	84.52	97.76	99.27	98.88
Grapes_untrained	68.73	85.80	76.03	82.11	75.33	89.11	84.32
Soil_vinyard_develop	97.76	99.87	98.85	97.81	82.35	98.50	98.52
Corn_senesced_green_weeds	89.12	71.48	80.84	65.53	51.68	82.48	94.39
Lettuce_romaine_4wk	91.72	89.04	94.94	80.89	36.98	86.05	90.26
Lettuce_romaine_5wk	94.81	98.81	99.68	67.28	39.54	97.41	99.38
Lettuce_romaine_6wk	93.05	19.54	97.71	67.25	25.54	96.94	91.16
Lettuce_romaine_7wk	93.32	93.08	94.11	39.25	39.53	92.99	91.49
Vinyard_untrained	57.43	45.36	52.63	49.36	41.62	43.28	70.89
Vinyard_vertical_trellis	98.23	98.01	88.60	41.75	77.70	83.45	98.17
OA	85.42±0.37	82.34±0.34	85.41±0.29	75.07±0.47	68.82±0.23	86.39±0.33	**91.45±0.21**
AA	91.46±0.25	81.16±0.21	90.88±0.31	71.34±0.35	61.99±0.19	89.74±0.24	**94.55±0.14**
Kappa	0.83±0.003	0.80±0.0.002	0.83±0.003	0.71±0.004	0.65±0.001	0.84±0.002	**0.90±0.001**

The best result are highlight in bold.

TABLE V

HSI CLASSIFICATION RESULTS (%) BY SELECTING 2% LABELED SAMPLES FOR EACH CLASS ON THE PAVIAU DATA SET

Land Cover Type	KNN	KPCA	RICA	LLE	LRR	SSAE	LRSSAE
Asphalt	88.98	82.79	89.67	81.92	85.75	88.96	91.93
Meadows	83.68	90.43	94.27	90.01	92.73	94.29	97.24
Gravel	62.36	27.68	64.22	34.53	36.45	73.41	73.65
Trees	89.85	81.46	81.49	57.83	73.34	87.07	93.79
Painted metal sheets	98.16	72.42	97.92	98.52	92.27	99.41	97.55
Bare Soil	74.40	61.07	54.82	69.63	36.57	61.94	85.66
Bitumen	62.14	34.14	11.80	86.53	1.88	73.83	76.32
Self-Blocking Bricks	74.89	90.14	74.63	88.50	60.05	79.46	87.91
Shadows	99.99	93.98	98.63	99.98	76.66	99.47	99.05
OA	82.34±0.33	79.81±0.47	82.48±0.34	81.58±0.23	74.89±0.26	86.48±0.36	**92.25±0.22**
AA	81.61±0.28	70.46±0.25	74.16±0.27	78.61±0.18	61.74±0.24	84.20±0.31	**89.24±0.23**
Kappa	0.76±0.003	0.73±0.003	0.76±0.002	0.75±0.001	0.65±0.0.002	0.82±0.003	**0.90±0.002**

The best result are highlight in bold.

TABLE VI

HSI CLASSIFICATION RESULTS (%) BY SELECTING 1% LABELED SAMPLES FOR EACH CLASS ON THE PAVIAC DATA SET

Land Cover Type	KNN	KPCA	RICA	LLE	LRR	SSAE	LRSSAE
Water	99.95	99.98	99.97	99.93	99.99	99.89	99.88
Trees	77.25	93.61	60.61	51.51	96.28	90.91	93.61
Asphalt	84.97	77.11	92.24	33.75	62.38	66.67	81.48
Self-Blocking Bricks	67.67	48.69	66.54	14.02	0.05	63.83	77.24
Bitumen	81.48	88.87	77.41	97.80	74.69	90.70	96.15
Tiles	82.76	97.99	94.57	89.29	97.36	95.38	94.69
Shadows	90.88	85.33	86.07	98.41	82.69	83.90	94.83
Meadows	96.14	82.51	95.27	87.69	68.57	92.88	98.62
Bare Soil	99.76	98.94	99.97	98.89	27.71	99.44	99.91
OA	94.11±0.26	95.45±0.42	93.67±0.34	91.84±0.21	91.17±0.38	95.38±0.24	**97.49±0.21**
AA	86.76±0.17	85.89±0.37	85.88±0.29	74.59±0.26	67.75±0.25	87.07±0.21	**92.93±0.19**
Kappa	0.89±0.001	0.92±0.004	0.89±0.0.003	0.85±0.002	0.84±0.003	0.92±0.002	**0.96±0.002**

The best result are highlight in bold.

learning, which needs to be constructed. The number of k must be specified both in the spatial and spectral graph, which are chosen from the set {3, 4, 5, 6, 7, 8, 9, 10}. In the spatial graph, required to calculate the distance between feature representations, the heat kernel parameters γ_1 are required, which can be selected from the set $\{10^{-9}, 10^{-8}, \ldots, 10^{8}, 10^{9}\}$. For the balancing parameters such as ξ_1 and ξ_2, which can be tuned from the set $\{10^{-9}, 10^{-8}, \ldots, 10^{8}, 10^{9}\}$.

C. Experimental Results

To evaluate the effectiveness of the proposed classification model, three evaluation indicators, the average accuracy (AA), overall accuracy (OA), and kappa coefficient (Kappa), are used in our experiment. Tables IV–VI record the average classification accuracies and the corresponding standard deviations of the seven algorithms, which runs over 10 independently.

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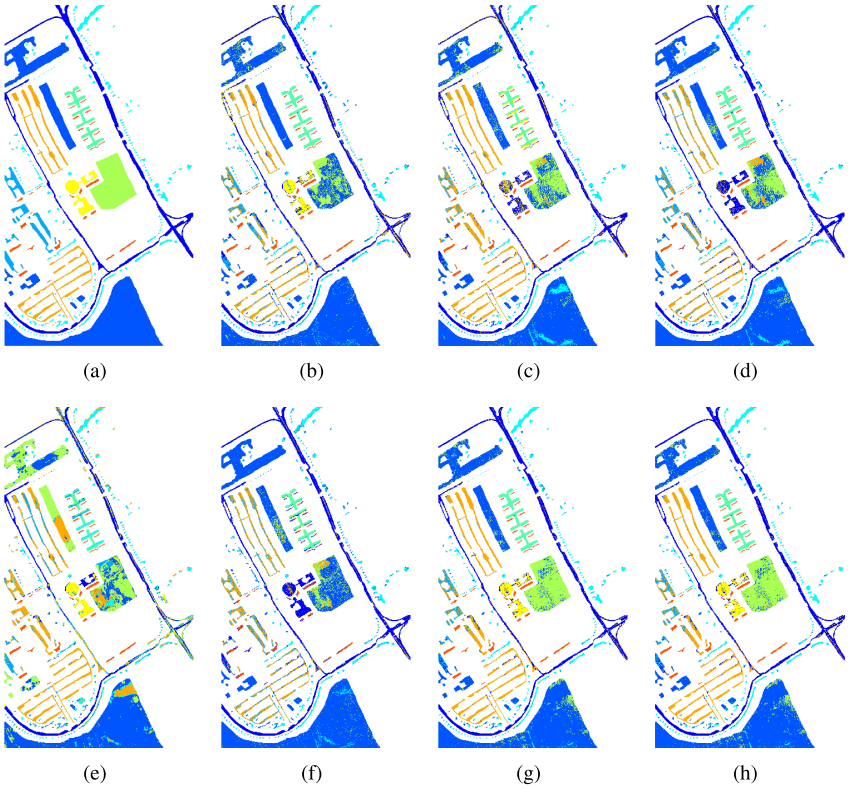

Fig. 6. Classification maps of the PaviaU data set by selecting 2% labeled samples for each class. Classification map obtained by (a) ground truth, (b) KNN, (c) KPCA, (d) RICA, (e) LLE, (f) LRR, (g) SSAE, and (h) LRSSAE.

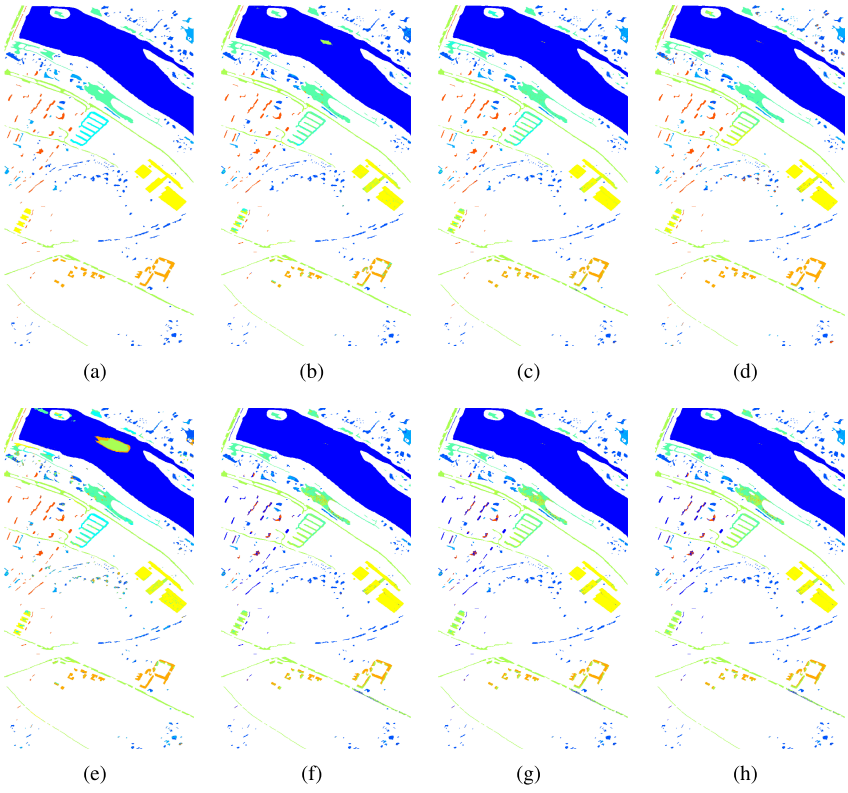

Fig. 7. Classification maps of the PaviaC data set by selecting 1% labeled samples for each class. Classification map obtained by (a) ground truth, (b) KNN, (c) KPCA, (d) RICA, (e) LLE, (f) LRR, (g) SSAE, and (h) LRSSAE.

From the results listed in Tables IV–VI and illustrated in Figs. 5–7, we have the following observations.

1) Compared with the basic SSAE, the LRSSAE achieves a much higher classification performance on three data

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TABLE VII
COMPARATIVE CLASSIFICATION RESULTS WITH BALANCED AND UNBALANCED TRAINING SAMPLES

Dataset		SSAE		LRSSAE	
		Balanced sample	Imbalanced sample	Balanced sample	Imbalanced sample
Salinas	OA(%)	86.39±0.33	86.72±0.41	91.45±0.21	90.30±0.48
	AA(%)	89.74±0.24	91.31±0.36	94.55±0.14	94.39±0.43
	Kappa	0.84±0.002	0.85±0.003	0.90±0.001	0.89±0.003
PaviaU	OA(%)	86.48±0.36	87.07±0.39	92.25±0.22	90.77±0.31
	AA(%)	84.20±0.31	82.55±0.23	89.23±0.23	87.77±0.26
	Kappa	0.82±0.003	0.82±0.003	0.90±0.002	0.88±0.002
PaviaC	OA(%)	95.38±0.24	95.82±0.34	97.49±0.21	97.29±0.23
	AA(%)	87.07±0.21	88.78±0.37	92.93±0.19	92.63±0.44
	Kappa	0.92±0.002	0.92±0.003	0.96±0.002	0.95±0.004

sets, which demonstrated that LRSSAE can learn more efficient and discriminative feature representations than SSAE. It also indicates that using the latent relationship learning is an effective way to improve feature learning, which is very important for HSI classification.

2) From Figs. 5–7, the comparison results with other methods, the LRSSAE, surpasses the other methods. It validates that the latent relationship learning and the superpixel constraint are the effective way to improving HSI classification accuracy.
3) Using the same labeled data to train the classifier, while the proposed method LRSSAE gets the best result among these compared approaches. Demonstrating that the more discriminative and representative feature representations are obtained by the LRSSAE model, in which the latent relationship guides the training of the SSAE model and then the constraint of latent relationship further enforce in superpixels.

D. Comparative Experiments With Balanced and Unbalanced Training Samples

To validate the performance of our method on the balanced and unbalanced training samples, we have added a comparative experiment of the balanced and unbalanced training samples on all data sets. We compare the proposed method with SSAE. In the experiment, 200 samples and 10% of the samples as the training samples of each class, which are randomly selected from the data sets, and remaining samples as the testing samples.

The classification performances of balanced and imbalanced samples are recorded in Table VII. The differences in the OA are very small both on the SSAE and LRSSAE method for the two different training samples. It demonstrates that our method is suitable for both balanced samples and imbalanced samples.

E. Comparison With the Other SSAE-Based Method

In this experiment, to validate the effectiveness of our method, we compared the LRSSAE with the representative SSAE-based method JS-SSAE [28] and the basic SSAE. The JS-SSAE jointly learns the spectral and spatial feature and extracts the spatial feature through the PCA transformation. We test different patch size from 3×3, 5×5, and 7×7, and then present the best result. The corresponding experimental results are shown in Table VIII. This shows that the

TABLE VIII
COMPARATIVE CLASSIFICATION RESULTS WITH SSAE-BASED METHODS

Dataset		SSAE	JS-SSAE	LRSSAE
Salinas	OA(%)	86.39±0.33	92.19±0.37	91.45±0.21
	AA(%)	89.74±0.24	95.13±0.19	94.55±0.14
	Kappa	0.84±0.002	0.91±0.002	0.90±0.001
PaviaU	OA(%)	86.48±0.36	89.76±0.28	92.25±0.22
	AA(%)	84.20±0.31	88.58±0.23	89.23±0.23
	Kappa	0.82±0.003	0.86±0.002	0.90±0.002
PaviaC	OA(%)	95.39±0.24	97.63±0.24	97.49±0.21
	AA(%)	87.07±0.21	93.90±0.17	92.93±0.19
	Kappa	0.92±0.002	0.96±0.001	0.96±0.002

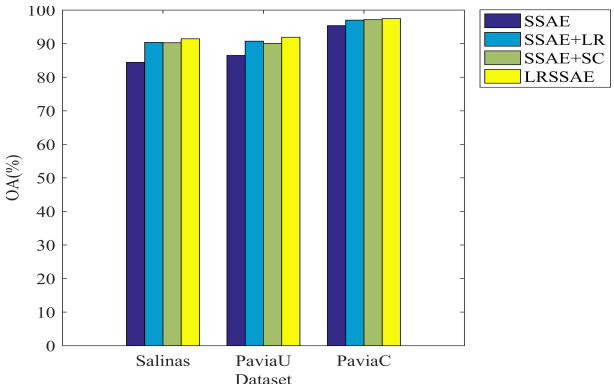

Fig. 8. Analyzing the importance of latent relationship and superpixel constraint.

JS-SSAE and LRSSAE can obtain a better performance in all data sets than basic SSAE; the differences in the OA of the JS-SSAE and LRSSAE are small in the Salinas and PaviaC data set; while the LRSSAE obtains improvement of 2.49% in the PaviaU data set compared to the JS-SSAE. The experimental results demonstrate that latent relationship learning is an efficient way of improving the feature learning ability.

F. Analyzing the Importance of Latent Relationship and Superpixel Constraint

In Fig. 8, we compared the proposed method LRSSAE with the other three variants of the SSAE model, including the basic SSAE, the SSAE-LR which embeds the latent relationship learning into the SSAE model, and the SSAE-SC which integrates the superpixel constraints into the SSAE model. Compared with the basic SSAE, we can see that the improvements in overall classification accuracy have been made in the SSAE-LR and SSAE-SC, which indicates that the latent relationship learning can enhance feature learning

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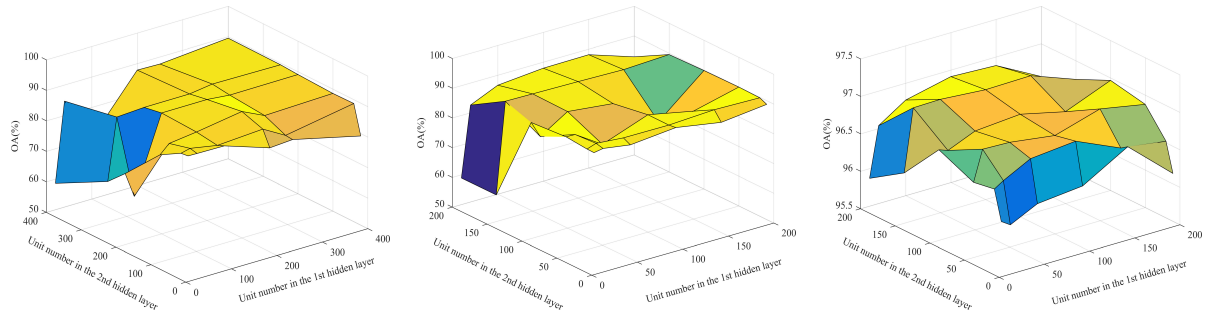

Fig. 9. Effect of structure parameters obtained by (a) Salinas data set, (b) PaviaU data set, and (c) PaviaC data set.

ability and the superpixel constraints are also an effective way to boost the classification performance. And these can also be validated by the result that the LRSSAE obtains the best classification performance compared with other methods.

G. Parameter Analysis

In LRSSAE, there are many parameters that need to be tuned in the training procedure, which is divided into two types: the hyperparameters and the structure parameters. Hyperparameters including the weight decay parameter μ, the sparse penalty parameter α, and the desired activation ρ. The values of these parameters are the same in diverse data sets, which are set to 1×10^{-4}, 3 and 0.01, respectively. There are also some other hyperparameters in those models, which need to tune in each data set, including learning rate and batch size. We adopt a learning rate decay way to train these model, such as for Salinas data set, the initial learning rate is 2×10^{-4}, and it is reduced by 0.96 every 100 iterations and the batch size is 32; while for the PaviaU and PaviaC data set, the initial learning rate is 2×10^{-3} and it is reduced by 0.96 every 500 iterations and the batch size is 32. There are two structure parameters: the number of hidden layers and the number of units in each hidden layer need to be tuned in the experiment to get a good performance.

We primarily discuss the influences of two structure parameters, including the number of hidden layers and the number of neurons in each hidden layer. Research works have demonstrated that the influences of the number of hidden units are more than those of the depth of network. Deep networks can get more abstract and high-order feature representation than the shallow model, but the computation complexity is growing simultaneously. Therefore, considering the trade-off between the computation complexity and classification accuracy, the LRSSAE model is set to two hidden layers. For the number of neurons in each hidden layers, an n-fold cross-validation is employed to find the optimal numbers [36]. We test how these parameters influence the performance of classification, set the number of neurons vary from n times of the input of each data set, n is set to [0.1, 0.2, 0.5, 1, 1.5, 2] in the first layer and second layer. Fig. 9 demonstrates that the influence of different hidden neurons for HSI classification by OA in diverse data sets. From Fig. 9, on the one hand, we can see that increase in the number of hidden layers can get a better performance among different hidden units indicating that a deep model can extract more efficient feature representation than shallow model. On the other hand, the combination of the number of units in the first and second layer influences the performance of classification, but the relationship between them is not linear simply. Thus, we choose those parameters according to the best performance of experimental results, n is set to 0.5 in the first layer in this experiment, 1 for the other.

V. Conclusion

In this article, we developed a novel unsupervised HSI classification method based on the SAE framework. The main contribution of this article is that a latent relationship is employed to guide the training procedure of SSAE model by adding the latent relationship-based graph regularization and superpixel constraints to the objective function. The LRSSAE model can help learn the accurate latent relationship and contextual information around pixels. The HSI classification results on three data sets demonstrated the effectiveness of our method. In the future work, the latent relationship can be integrated into the new DLMs, such as generative adversarial nets (GANs) and graph convolutional network (GCN), which preserve the original spectral–spatial correlation.

Appendix

A. Optimization for θ

θ is a collection of $\mathbf{W}$ and $\mathbf{b}$. We calculate the optimization procedure of $\mathbf{W}$ and $\mathbf{b}$ severally, which can be performed via the stochastic gradient descent and back-propagation algorithm, iteratively.

In (14), the objective function can be divided into four parts: the SAE, latent relationship, superpixel constraint, and a linearized penalty term of the augmented Lagrangian function. We calculate them as follows.

$$\begin{aligned}\min_{\theta} J(\theta) = \min_{\theta} \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2} \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 \right) \\ + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)} \right)^2 \\ + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho \| \hat{\rho}_j) + \xi_1 \operatorname{tr}(\mathbf{a}\mathbf{L}\mathbf{a}^T) \\ + \xi_2 \sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2} \right) \\ + \frac{\tau}{2} \left\| \mathbf{U}(\theta) - \left(\mathbf{R} + \frac{\omega}{\tau} \right) \right\|_F^2 .\end{aligned}$$

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The first part is the SAE, the objective function is defined in (3)

$$\begin{aligned}\min_{\theta} J_{\mathrm{SAE}}(\theta) = \min_{\theta} \frac{1}{\mathrm{M}} \sum_{i=1}^{M} \left(\frac{1}{2} \|\mathbf{y}_i - \mathbf{x}_i\|_2^2 \right) \\ + \frac{\mu}{2} \sum_{l=1}^{2} \sum_{i=1}^{S^{(l)}} \sum_{j=1}^{S^{(l+1)}} \left(\mathbf{W}_{ij}{}^{(l)} \right)^2 \\ + \alpha \sum_{j=1}^{S} \mathrm{KL}(\rho \| \hat{\rho}_j).\end{aligned}$$

The gradient of the objective function J_{SAE} with respect to $\mathbf{W}$ and $\mathbf{b}$ can be computed as follows:

$$\begin{cases} \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \dfrac{\partial J_{SAE}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \mathbf{a}_j{}^{(l-1)} \delta_i{}^{(l)} \\ \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{b}_i^{(l)}} = \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{b}_i{}^{(l)}} = \delta_i{}^{(l)} \end{cases}$$

$$\begin{cases} \delta_i{}^{(2)} = \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{z}_i{}^{(2)}} = -(\mathbf{x} - \mathbf{y}) f'\left(\mathbf{z}_i{}^{(2)}\right) \\ \delta_j{}^{(1)} = \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{z}_j{}^{(1)}} \\ \qquad = \left[\left(\sum_{i=1}^{M} \mathbf{W}_{ij}^{(2)} \delta_i{}^{(2)} \right) + \beta \left(-\dfrac{\rho}{\hat{\rho}_j} + \dfrac{1-\rho}{1-\hat{\rho}_j} \right) \right] \mathbf{f}'\left(\mathbf{z}_j{}^{(1)}\right) \end{cases}$$

where f' denotes the inverse function of $f(\cdot)$, l denotes the number of layer in single SAE, $\mathbf{x}$ and $\mathbf{y}$ represent the input vector and reconstruction vector of the input.

The second term is the penalty of latent relationship, which is defined in (10)

$$\begin{aligned}\min J_{\mathrm{LR}}(\theta) &= \frac{1}{2} \sum_{i,j}^{S} \mathbf{R} \|\mathbf{a}_i - \mathbf{a}_j\|_F^2 \\ &= \mathrm{tr}(\mathbf{a}\mathbf{L}\mathbf{a}^T).\end{aligned}$$

The gradient of the objective function J_{LR} with respect to $\mathbf{W}$, $\mathbf{b}$ can be computed as follows:

$$\begin{cases} \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \mathbf{a}_j{}^{(l-1)} \cdot \delta'_i{}^{(l)} \\ \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{b}_i{}^{(l)}} = \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{b}_i{}^{(l)}} = \delta'_i{}^{(l)} \end{cases}$$

$$\begin{cases} \delta'_i{}^{(2)} = \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{z}_i{}^{(2)}} = 0 \\ \delta'_j{}^{(1)} = \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{z}_j{}^{(1)}} \\ \qquad = \mathbf{R} \sum_{i,j}^{S} \left(\left(\mathbf{a}_i{}^{(1)} - \mathbf{a}_j{}^{(1)} v \right) \left(f'\left(\mathbf{z}_i{}^{(1)}\right) - f'\left(\mathbf{z}_j{}^{(1)}\right) \right) \right). \end{cases}$$

Similarly, the optimization problem of the superpixels constraint is defined in (11)

$$\begin{aligned}\min J_{\mathrm{SP}}(\theta) &= \sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2} \right) + \varphi \|\mathbf{R} - \bar{\mathbf{R}}\|_F^2 \\ &\Leftrightarrow \min J_{\mathrm{SP}}(\theta) = \sum_{j=1}^{S} \exp\left(\frac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2} \right)\end{aligned}$$

The gradient of the objective function J_{SP} with respect to $\mathbf{W}$, $\mathbf{b}$ can be computed as follows:

$$\begin{cases} \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \mathbf{a}_j{}^{(l-1)} \delta''^{(l)}_i \\ \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{b}_i{}^{(l)}} = \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{b}_i{}^{(l)}} = \delta''^{(l)}_i \end{cases}$$

$$\begin{cases} \delta''^{(2)}_i = \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{z}_i{}^{(2)}} = 0 \\ \delta''^{(1)}_j = \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{z}_j{}^{(1)}} \\ \qquad = \sum_{j}^{S} \left(2(\mathbf{a}_j - \varsigma_k) \exp\left(\dfrac{\|\mathbf{a}_j - \varsigma_k\|_2^2}{\gamma_2} \right) f'\left(\mathbf{z}_j{}^{(1)}\right) \right). \end{cases}$$

Finally, a linearized penalty term of the augmented Lagrangian function, which is denoted by J_{L}, can be defined as follows:

$$\min J_L(\theta) = \frac{\tau}{2} \left\| \mathbf{U}(\theta) - \left(\mathbf{R} + \frac{\omega}{\tau} \right) \right\|_F^2.$$

The derivatives of J_{L} and the parameters can be calculated as follows:

$$\begin{cases} \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{W}_{ij}{}^{(l)}} = \mathbf{a}_j{}^{(l-1)} \delta'''^{(l)}_i \\ \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{b}_i{}^{(l)}} = \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{z}_i{}^{(l)}} \dfrac{\partial \mathbf{z}_i{}^{(l)}}{\partial \mathbf{b}_i{}^{(l)}} = \delta'''^{(l)}_i \end{cases}$$

$$\begin{cases} \delta''^{(2)}_i = \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{z}_i{}^{(2)}} = 0 \\ \delta''^{(1)}_j = \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{z}_j{}^{(1)}} \\ \qquad = \tau \left(\mathbf{U}(\theta) - \left(\mathbf{R} + \dfrac{\omega}{\tau} \right) \right) \Delta \end{cases}$$

$$\Delta(i,j) = \frac{\partial \mathbf{U}}{\partial \mathbf{z}} = \begin{cases} -2(\mathbf{a}_i - \mathbf{a}_j) \exp\left(-\mathbf{r}_{ij}{}^2\right) \left(f'\left(\mathbf{z}_i{}^{(1)}\right) - f'\left(\mathbf{z}_j{}^{(1)}\right) \right) \\ \text{where } x_i \in \mathcal{N}_{k_1}(x_j) \text{ or } x_j \in \mathcal{N}_{k_1}(x_i) \\ 0, \quad \text{otherwise} \end{cases}$$

where Δ denotes the gradient of spectral–spatial regularization $\mathbf{U}$ with $\mathbf{z}$, whereas the latent relationship of spatial correlation is independent of $\mathbf{z}$, so Δ represents only the derivatives of the spectral information.

The gradient of the objective function J with respect to $\mathbf{W}$ and $\mathbf{b}$ can be obtained by combining the gradients of the two terms

$$\begin{cases} \dfrac{\partial J}{\partial \mathbf{W}_{ij}{}^{(l)}} = \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{W}_{ij}{}^{(l)}} + \xi_1 \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{W}_{ij}{}^{(l)}} + \xi_2 \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{W}_{ij}{}^{(l)}} + \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{W}_{ij}{}^{(l)}} \\ \dfrac{\partial J}{\partial \mathbf{b}_i^{(l)}} = \dfrac{\partial J_{\mathrm{SAE}}}{\partial \mathbf{b}_i^{(l)}} + \xi_1 \dfrac{\partial J_{\mathrm{LR}}}{\partial \mathbf{b}_i^{(l)}} + \xi_2 \dfrac{\partial J_{\mathrm{SP}}}{\partial \mathbf{b}_i{}^{(l)}} + \dfrac{\partial J_{\mathrm{L}}}{\partial \mathbf{b}_i{}^{(l)}} \end{cases}$$

where the third term represents the derivatives of the superpixels constraint and parameters, which is employed in the last layer of LRSSAE, so it is equal to 0 in other layers of the LRSSAE.

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Stochastic gradient descent method is employed to update the parameters $\mathbf{W}$ and $\mathbf{b}$ of the SAE

$$\begin{cases} \mathbf{W}^{(l)} = \mathbf{W}^{(l)} - \varepsilon_1 \left[\dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \dfrac{\partial J}{\partial \mathbf{W}_{ij}{}^{(l)}} + \mu \mathbf{W}^{(l)} \right] \\ \mathbf{b}^{(l)} = \mathbf{b}^{(l)} - \varepsilon_1 \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \dfrac{\partial J}{\partial \mathbf{b}_i{}^{(l)}} \end{cases}$$

where ε_1 denotes the learning rate.

B. Optimization for $\boldsymbol{R}$

$\mathbf{R}$ is solved when θ are fixed. The optimization function defined in (15) is expressed as follows:

$$\begin{aligned} \min_{\mathbf{R}} J(\mathbf{R}) &= \min_{\mathbf{R}} \xi_1 \operatorname{tr}(\mathbf{aL(a)}^T + \xi_2 \varphi \|\mathbf{R} - \bar{\mathbf{R}}\|_F^2 \\ &\quad + \frac{\tau}{2} \|\mathbf{R} - \left(\mathbf{U}(\theta) - \frac{\omega}{\tau} \right) \|_F^2 \\ &= \min_{\mathbf{R}} \frac{1}{2} \xi_1 \sum_{i,j}^{S} \mathbf{R} \|\mathbf{a}_i - \mathbf{a}_j\|_2^2 + \xi_2 \varphi \|\mathbf{R} - \bar{\mathbf{R}}\|_F^2 \\ &\quad + \frac{\tau}{2} \left\| \mathbf{R} - \left(\mathbf{U}(\theta) - \frac{\omega}{\tau} \right) \right\|_F^2 . \end{aligned}$$

The derivative with respect to $\mathbf{R}$ can be computed by the following:

$$\begin{aligned} \frac{\partial J(\mathbf{R})}{\partial \mathbf{R}} = \frac{\xi_1}{2} \sum_{i,j}^{S} \|\mathbf{a}_i - \mathbf{a}_j\|_2^2 + 2\xi_2 \varphi (\mathbf{R} - \bar{\mathbf{R}}) \\ + \tau \left(\mathbf{R} - \left(\mathbf{U}(\theta) - \frac{\omega}{\tau} \right) \right) . \end{aligned}$$

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